This document provides a high-level view of the information technology infrastructure in place at EnCap Investments. Detailed documents providing specifications are also available upon request.

Information Technology

Overview

EnCap Investments L.P.

Updated: Jan 2024
By: Shamil Zuberi

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Introduction

EnCap Investments uses state of the art data center equipment at the Houston Office in addition to cloud-hosted services which serves as the secondary data center. We continue to update our equipment to keep these services secure and highly available, with several levels of functional redundancy.

EnCap Investment's IT infrastructure employs the industry standard Microsoft Windows Server platform, and all internal workstations use Windows 10 Pro.

Network Infrastructure

Wide Area Network

The Houston Office has dual firewalls (in High Availability) and dual circuits in place with automatic failover, should one go down or experience any issue/outage. Cloud Infrastructure is connected through two independent VPN tunnels to the Houston Office.

Local Area Network

The local area networks use gigabit ethernet connectivity between hosts via Cisco POE switches. Network segregation is used within the LAN to separate different types of traffic. For example: subnet for servers, workstations, phone, WIFI, VPN, etc. Wireless connectivity to the network is securely available only with RADIUS authentication.

Windows Server Infrastructure

The server infrastructure is based on the Microsoft Windows Server 2016 Active Directory (AD) Domain. There are two sites in the Forest, one in Houston and the second in Microsoft Azure. Multiple servers provide authentication, file storage, printing, and line of business application needs. Domain authentication is possible from either of the two locations.

Use of DFS-R for File Data Replication

A Distributed File System with Replication (DFS-R) has been implemented in the environment for file and folder redundancy. DFS-R technologies offer Wide Area Network (WAN) replication as well as simplified, highly available access to geographically dispersed data. Functionally, files and data generated at either the Houston or Azure locations are replicated to each other in real time.

Group Policies

Enterprise-wide group policies are configured to secure access to data and network resources.

Email Infrastructure

EnCap Investments uses the Microsoft O365 platform providing mail and app availability at 99.99%.

Mobile Device Management

EnCap employs Microsoft's Intune mobile device management (MDM) tool to manage all mobile devices, providing secure control over dispersed company information.

Email Spam Filtering and Advanced Threat Management

Microsoft's Active Thread Protection (ATP) protects against various inbound spam, malware, and virus attacks, providing zero-day protection to safeguard internal systems. Incoming messages and attachments are processed by machine learning that can detect and quarantine mail with malicious intent.

Remote Access (VPN & MFA)

Virtual Private Network (VPN) based remote access is done using the Cisco AnyConnect infrastructure. Redundancy is achieved with two ingress points, one each in Houston and Microsoft Azure. Multi-Factor Authentication (MFA) for all VPN connections are available using DUO Mobile's MFA product.

Anti-Virus and End Point Protection

EnCap uses the Sophos Central cloud based anti-virus/anti-spam solution to keep servers and workstations protected from known and zero-day threats.

A second end point protection platform from Carbon Black is installed on all servers and workstations to filter traffic and monitor activity. Carbon Black can lock down an endpoint in case it identifies anything to be malicious.

Backups

EnCap Investments uses several platforms for data backup.

Barracuda Networks

A daily backup is performed with this on-site, disk-to-disk appliance. For redundancy, these backups are saved to the cloud.

Veeam Backup & Recovery

Veeam backups are configured to perform multiple server backups and snapshots daily. This allows for faster recovery of virtual machines. Veeam provides automatic recovery and testing of every backup and every replica. Veeam solution also serves as a Disaster Recovery (DR) solution as it saves full backups of all data to the SAN.

Microsoft Azure File Backup

Files are backed up on Azure and configured to keep 6 copies across data centers in the US in case of failure at any site.

Web Traffic Filtering

EnCap Investments uses Sophos' web filter to centrally manage web traffic and content at all endpoints. The Sophos web filter allows access to online apps and tools without exposure to web-borne malware and viruses. As a comprehensive solution for web security and content filtering, it unites spyware, malware, and virus protection with a powerful, configurable policy and reporting engine. Advanced features allow for adaptation to emerging requirements such as social-network regulation, remote content filtering, and visibility into SSL-encrypted traffic.

Disaster Recovery, Redundancy & Business Continuity

EnCap Investment's IT infrastructure is highly redundant by virtue of the replication at Houston, and multiple geo location copies using Microsoft Azure. EnCap maintains a list of business-critical apps and data and replicates these in real-time across sites. If a location is lost or unavailable due to disaster, end-user connections to another site are seamless. Failover testing is done quarterly.

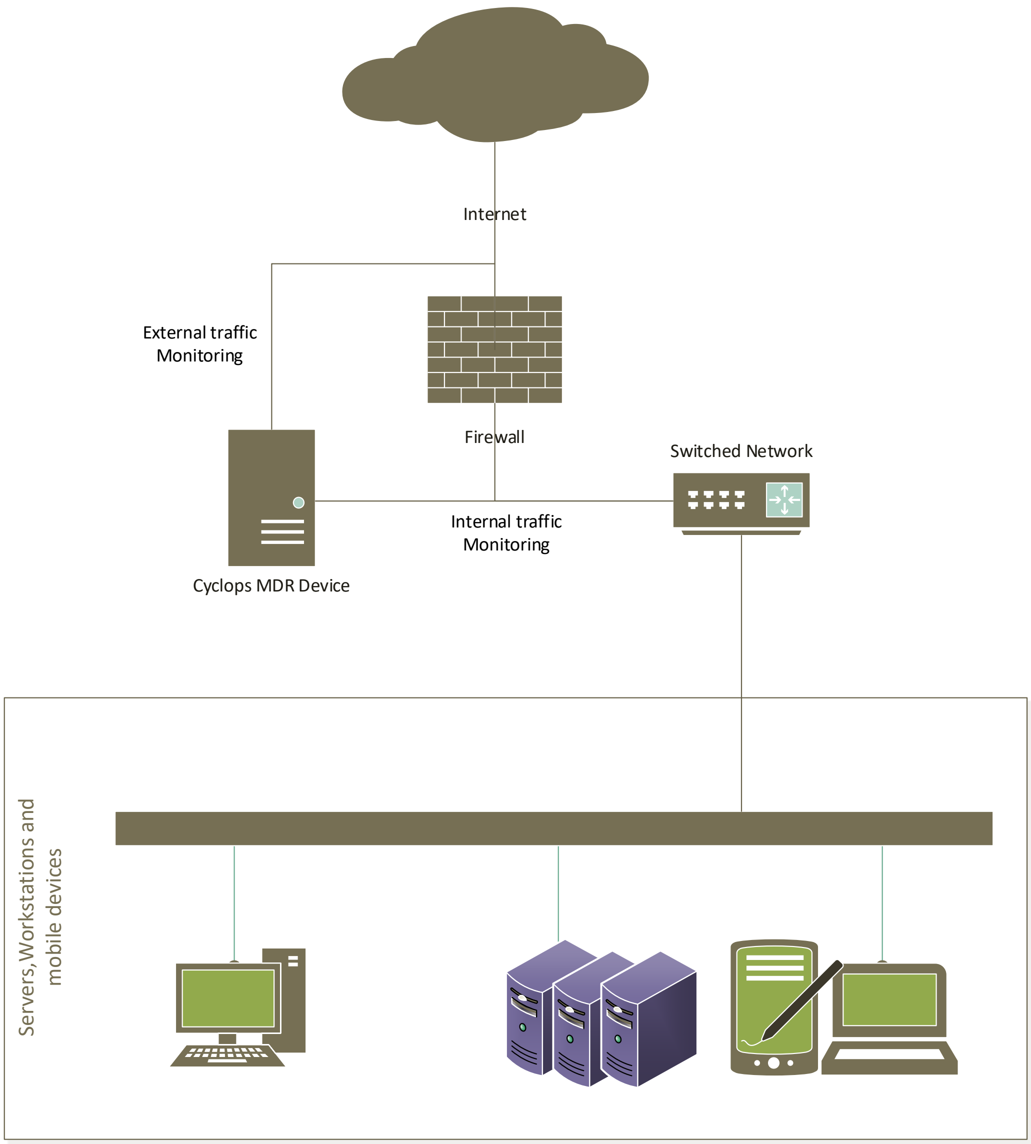
Cybersecurity

EnCap Investments has implemented several technologies to enhance cybersecurity for production systems.

Managed Detection and Response (MDR)

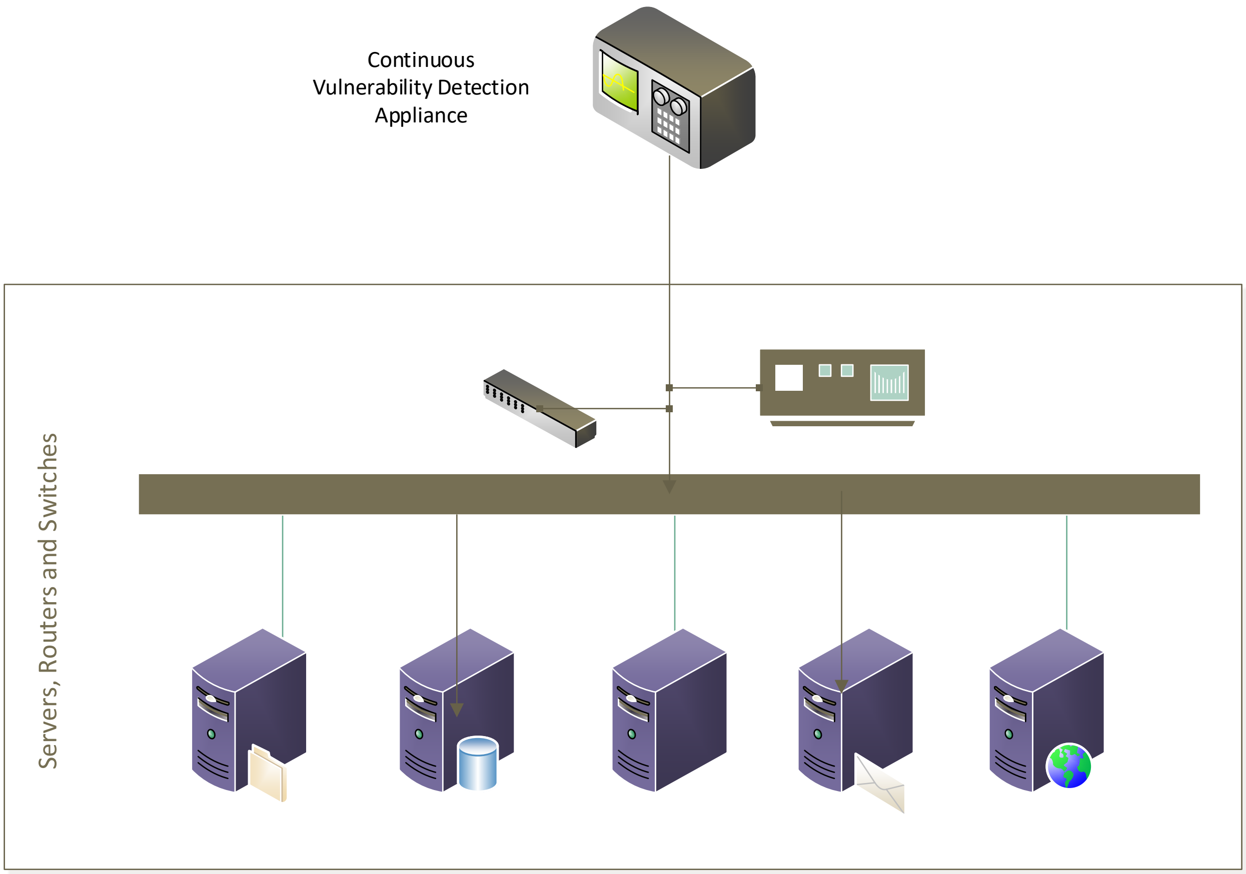
Devices from eSentire constantly listen to the traffic both inside and outside of the network focusing on hunting and detecting previously undetected threats that might bypass other security controls and contain them before they impact business.

This real time monitoring, detection, and prevention ensures an extra layer of safety and security for the data and computing environment.



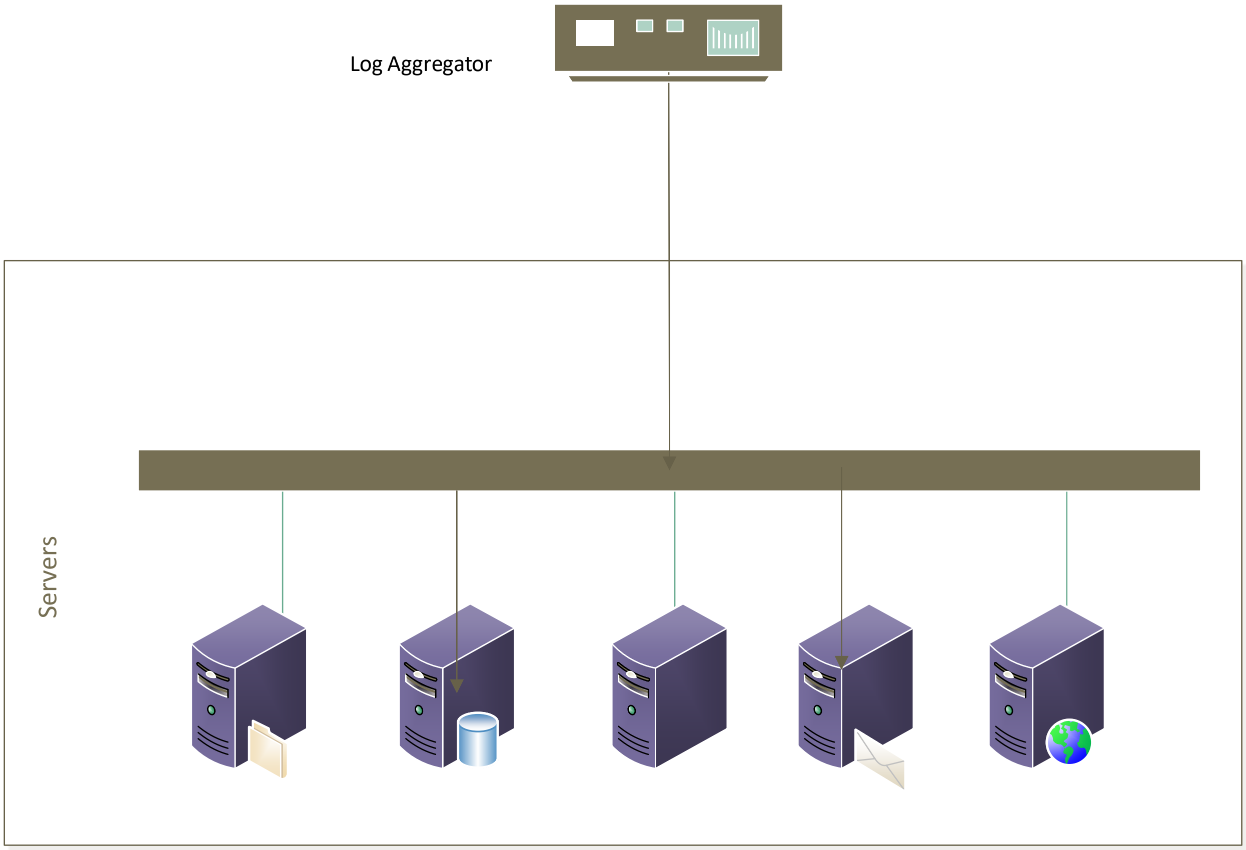
Continuous Vulnerability Detection (CVS) using Tenable

Networks without a CVS solution can suffer from patching and device firmware vulnerabilities. The latency between the discovery of a vulnerability and when it can be eliminated through patching from a vendor is usually high. EnCap Investments has implemented a CVS solution providing continuous scanning for vulnerabilities of network devices, servers, and workstations. CVS also provides for the scanning of web applications and other software solutions that rely on industry standard software engines like SQL and CRM systems, so the patch management latency window shrinks to almost zero and enhances the exploit resiliency of systems.



Log Aggregation using Sumo Logic

The ability to detect and respond to server issues is directly related to the uptime of the server computing environment. EnCap Investments has deployed a log aggregation tool that monitors logs from all critical and non-critical servers and alerts the systems team of issues detected in the server logs. This allows for an instantaneous response to issues that may not be detected in a timely fashion otherwise. Log aggregation is an added component that lends to a highly available computing environment.


Log Aggregator
Servers